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Task Description:	The overall objective of this NASA Specialized Center of Research (NSCOR) is to model the relative risk of colonic and stomach tumorigenesis for high priority space radiation beams and compare to gamma radiation where human epidemiologic data are available. It will rely primarily on mouse models of gastrointestinal (GI) cancer and human colonic epithelial cell (HCEC) approaches that have been optimized for low dose exposures. This will involve quantitative and qualitative analyses of tumors as well as normal GI epithelium after radiation. Results will be extrapolated to radiation-induced molecular changes in state-of-the-art 2D and 3D cultured human colonocytes. While limited studies at select loci, such as HRPT, have shown HZE-induced mutations including splicing errors, frameshifts, and complex changes, we will carry out genome-wide analysis using the power of modern nextgen sequencing (NGS) to assess the spectrum of mutations in radiation-induced tumors, and also persistent genome-wide epigenomic changes in normal cells after ionizing radiation (IR). These NGS omics findings will be integrated with transcriptomics and metabolomics results as well as events at the protein and cellular levels to develop a systems biology model for GI tumorigenesis. By assessing both quantitative in vivo tumorigenesis data and molecular events in mouse tissues and human colonocytes, space radiation-induced tumorigenesis will be compared to human epidemiologic data for low LET (linear energy transfer) radiation exposures, and approaches will be developed to model human risks for GI cancer by space radiation. We continue to employ two models of GI cancers: the APC1638N model, which was generated by a targeted modification of the Apc gene at a position corresponding to aa1638. These mice develop very few adenomas in the small as well as in the large intestine but post-radiation tumor numbers show marked increases. Radiation induced substantial gastric tumorigenesis as well in this model. The second model is the CDX2P Apc flox/+ that has one APC allele inactivated almost exclusively in the colon leading to adenomas and subsequent a low rate of adenocarcinomas in the colon that is increased post-irradiation. Data being collected from these two relevant mouse models will be used to calculate risk estimates for NASA. Since low-LET risk estimates are available for human populations, we will then be able to model the relative risk for space radiation. In particular, this ratio can then be extrapolated to human exposure to low-LET radiation to then make an estimate of risk for colorectal cancer in humans exposed to space radiation compared to the standard risks based on atomic bomb survivors and other exposure populations. In order to answer the critical question of risk for colonic tumorigenesis after acute and chronic space radiation exposures, in vitro human cell culture and animal models for CRC initiation and progression are needed. We are investigating the effects of high-LET particle and high-energy protons in colonic epithelial cells with respect to the nature of damage and consequent tumor induction using biological cancer-related endpoints and systems biology (genomics, proteomics, and metabolomics) approaches. In addition, we propose to compare these results to the biological consequences of low dose rates and mixed fields of radiation which astronauts are likely to be continuously exposed. The studies of low- and high-LET radiation on colonocytes gain more relevance with increasing age when premalignant adenomas and other colonic lesions such as flat lesions and microscopic aberrant crypt foci become more prevalent. Pre-existing undetectable genetic or chromosomal aberrations in astronauts in addition to de novo mutations resulting from long-term irradiation may increase the chances of colonic stem cell transformation. Plans for this proposal build on our past accomplishments with GI tumorigenesis models and are focused on studies at low doses with an expanded range of NASA's high priority beams. This proposal is also focused on assessing the effects of gender, dose-rate, age, and genetic heterogeneity in mouse models. Molecular events triggered by space radiation will be investigated in depth and the mechanistic basis for persistent aberrant cell-signaling after HZE radiation will be dissected. Genome-wide NGS combined with other omics approaches will contribute to a systems biology understanding of the effects of space radiation in the intestines and stomach. Both quantitative murine tumor data and results for underlying molecular events in mouse and human GI cells will contribute to modeling of human risks for GI cancer by space travel. The overall aims of this NSCOR are summarized below. Aim 1 (Project 1). Quantitatively assess GI tumorigenesis in mouse models of GI cancer and collect samples for qualitative analysis. Aim 2 (Project 2). Dissection of the signaling events and consequences in GI cells of the persistent effects of space radiation. Aim 3 (Project 3): Characterization of IR-induced neoplastic events in normal diploid human colonocytes. Aim 4 (Project 4). Development of systems biology (Project 4A) and mathematical modeling (Project 4B) approaches for GI cancer risk assessment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Gastrointestinal tumors are frequent in the U.S, and American Cancer Society estimates that there will be 132,700 new cases of colorectal cancer (CRC) with 49,700 persons predicted to die of the disease in 2015. Colorectal cancer when considered in men and women is the second leading cancer killer in the United States (http://www.cdc.gov/). Considering the high frequency of colorectal tumors in the American population, an even small increase by space radiation could have a major impact on risk estimates and planning of future manned space missions. The effect of complex DNA lesions, which are produced by HZE radiation, on intestinal tumorigenesis may provide insight into mutagenic processes affecting genome integrity and colorectal carcinogenesis.
	Considering the increased colorectal cancer incidence in A-bomb survivors and the novel characteristics of space radiation, cancer causation by space radiation could potentially be even greater. One general approach for risk assessment is to determine the relative biologic effectiveness (RBE) of various parameters for space radiation compared to terrestrial radiation exposures. For HZE ions there is more uncertainty with RBE for cellular parameters, such as chromosome aberrations and cell lethality, where values obtained from in vitro studies typically ranges from 2 to 11-fold. However, we do not have sufficient in vivo mechanistic data in human or in animal models representing the human disease to develop reliable risk prediction models for space radiation-induced colorectal cancer (CRC). Considering the limitation of acquiring human data, our study is focused on acquiring sufficient data in mouse models of human CRC with an aim to aid in risk model development for safe human space exploration. Task progress on GI tumorigenesis (GU). Since the award of the this NSCOR in April 2015, we have participated in one beam run (Fall 2015) at the NASA Space Radiation Laboratory (NSRL) and currently are scaling up APC1638N/+ mice breeding in preparation for our spring 2016 beam run. We have already demonstrated higher tumorigenesis with space

Task Progress:	radiation relative to gamma rays as well as important qualitative differences in the IR effects in both normal GI epithelial cells and GI tumors. The goals going forward are to assess tumorigenesis at more relevant lower doses and varied dose rates with a broader range of ions. Since maximal effects have been seen with 28-Si ions in our previous studies, we have continued experiments with 28-Si (300 MeV/n) albeit at new lower doses (5 and 1 cGy). We have also started studies using 16-O (325 MeV/n) and 4-He (250 MeV/n) ions, which are NASA prioritized ions of lower Z value proposed in this NSCOR. Additional mice will be irradiated in spring 2016 to fulfill statistical requirements. Apart from acute exposures (dose rate 10 to 50 cGy/min), we have also initiated studies using low dose rate (0.33 cGy/min) exposures and will continue in spring 2016. For comparison, we have exposed mice to gamma radiation at doses equal to HZE ions. Mice irradiated in 2016 fall will be ready for data collection 150 days after radiation exposure. Alongside, we have started backcrossing APC1638N/+ of C57BL/6J background into C3H/HeN and BALB/c background for the genetic heterogeneity study proposed in the project and is expected to take about a year to fully establish the model in these backgrounds. Proton radiotherapy induces more precise DNA damage at the tumor site with reduced side effects to adjacent normal tissues. However, protons emerging from solar flares are considered to be one of the major acute risk factors for astronauts. Even though some shielding is possible, the long-term biological effects of proton irradiation in cancer initiation and progression compared with conventional photon irradiation remain poorly characterized. In this series of studies, using a human familial adenomatous polyposis syndrome susceptible mouse model, we show that whole-body irradiation with simulated solar particle event (SPE) protons are more effective in inducing senescence-associated inflammatory responses (SIRs), which are involved in colon cancer initiation and progression. After simulated SPE irradiation, a subset of SIR genes (Troy, Sox17, Opg, Faim2, Lpo, Tlr2, and Ptges) and a gene known to be involved in invasiveness (Plat), along with the senescence-associated gene (P19Arf) are markedly increased. Following these changes, loss of Casein kinase Ia (CKIa) and induction of chronic DNA damage and TP53 mutations are increased compared with the same doses of acute proton or x-ray irradiation. Simulated SPE irradiation (2 Gy, 92% 50 MeV/n, 8% 60-150 MeV/n) also increases the number of colonic polyps, and invasive adenocarcinomas compared to 2 Gy (50 MeV/n) single dose proton or 2 Gy (250 kVp) x-ray. Thus, exposure to SPE irradiation elicits significant changes in colorectal cancer initiation and progression Considering that the responses were more pronounced after HZE irradiation relative to gamma radiation and 56-Fe and 28-Si were tested at moderately high doses in previous studies, the current project will expand persistent effect studies to additional priority beams, and lower radiation doses and dose rates more relevant to the space environment. To this end, we have irradiated male C57BL/6J and Lgr5-EGFP-IRES-creERT2 (Lgr5 mice) during 2015 fall run at 5 and 10 cGy and samples collected 2 months after exposure for molecular analysis, which is ongoing. Later, samples will also be collected 12 months after radiation to determine persistence of effects. The goal is to characterize persistent effects of space radiation at low doses and dose rates on GI epithelial and stem cells. Irradiation of mice is performed concurrently with the Project 1 mice. Work is also ongoing on human colonic epithelial cells (HCECs) with plans to irradiate these during the 2016 beam runs at NSRL. The work on the genomic project has been started with tumor samples from a moderate dose (1.6 Gy) of 56-Fe radiation archived from previous NSCOR. The goal is to standardize the methodology for signal detection using a relatively high dose of HZE radiation before analyzing lower doses (<1 Gy) of HZE radiation. Additionally, tumorigenesis data from the previous NSCOR is also used to calculate relative biological effectiveness (RBE) and develop modeling approach for colorectal cancer risk estimates in astronauts.

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